



MediQuick AI

An Advanced Multimodal Medical Triage Assistant

Mid-Semester Progress Review · June 2026



The Problem

*“How urgent is this — **and who should I see?**”*

1

Noise, not clarity

Search results are overwhelming and often alarming, not reassuring.

2

Cost & access

Booking a consultation for every minor concern is impractical and expensive.

3

Wrong question

People need to know the urgency and the right specialist — not a self-diagnosis.



The Approach

MediQuick AI triages — it does not diagnose. Describe symptoms in chat, optionally add a photo, and receive a structured, urgency-ranked assessment.

Multimodal

Understands text and images together via a Vision-Language Model.

Structured JSON

Every reply follows a strict, validated six-field schema.

Three urgency tiers

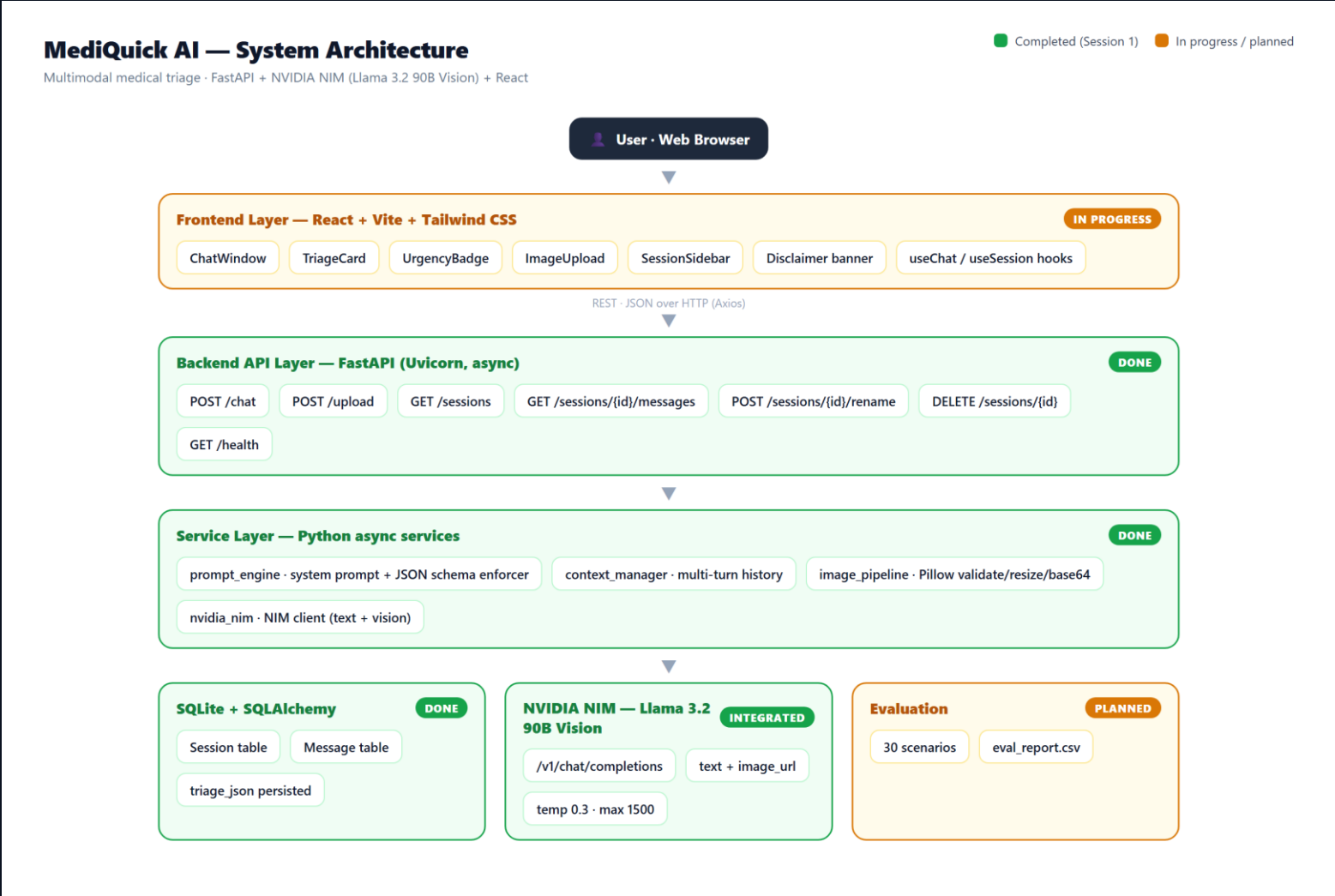
Self-care · consult a doctor · seek emergency care.

Safe by design

Emergency override and a disclaimer on every response.



System Architecture



Green = complete and validated this milestone · Amber = planned for the next phase

+ Demo • Symptom Chat

1

User types a symptom in plain language.

2

Backend adds the system prompt and prior turns.

3

The model returns a structured triage card.

⚠️ MediQuick AI provides preliminary triage guidance only. It is not a substitute for professional medical advice, diagnosis, or treatment.

MediQuick AI
Triage Assistant

+ New Conversation

RECENT

- Red circular rash on arm
Today · 2 messages
- Persistent headache 4 days
Today · 4 messages
- Sprained ankle after running
Yesterday · 6 messages
- Sore throat and mild fever
18 Jun · 3 messages

Red circular rash on arm Llama 3.2 Vision · NVIDIA NIM

I have a red circular rash on my arm for 3 days, it is slightly itchy with a clearer center.

+ POSSIBLE CONDITIONS

Ringworm Contact dermatitis Eczema

URGENCY

⚠️ See a Doctor Soon

The rash has been present for 3 days and is itchy, which may require further evaluation and guidance from a doctor.

RECOMMENDED SPECIALIST

Dermatologist

SUGGESTED FOLLOW-UP QUESTIONS

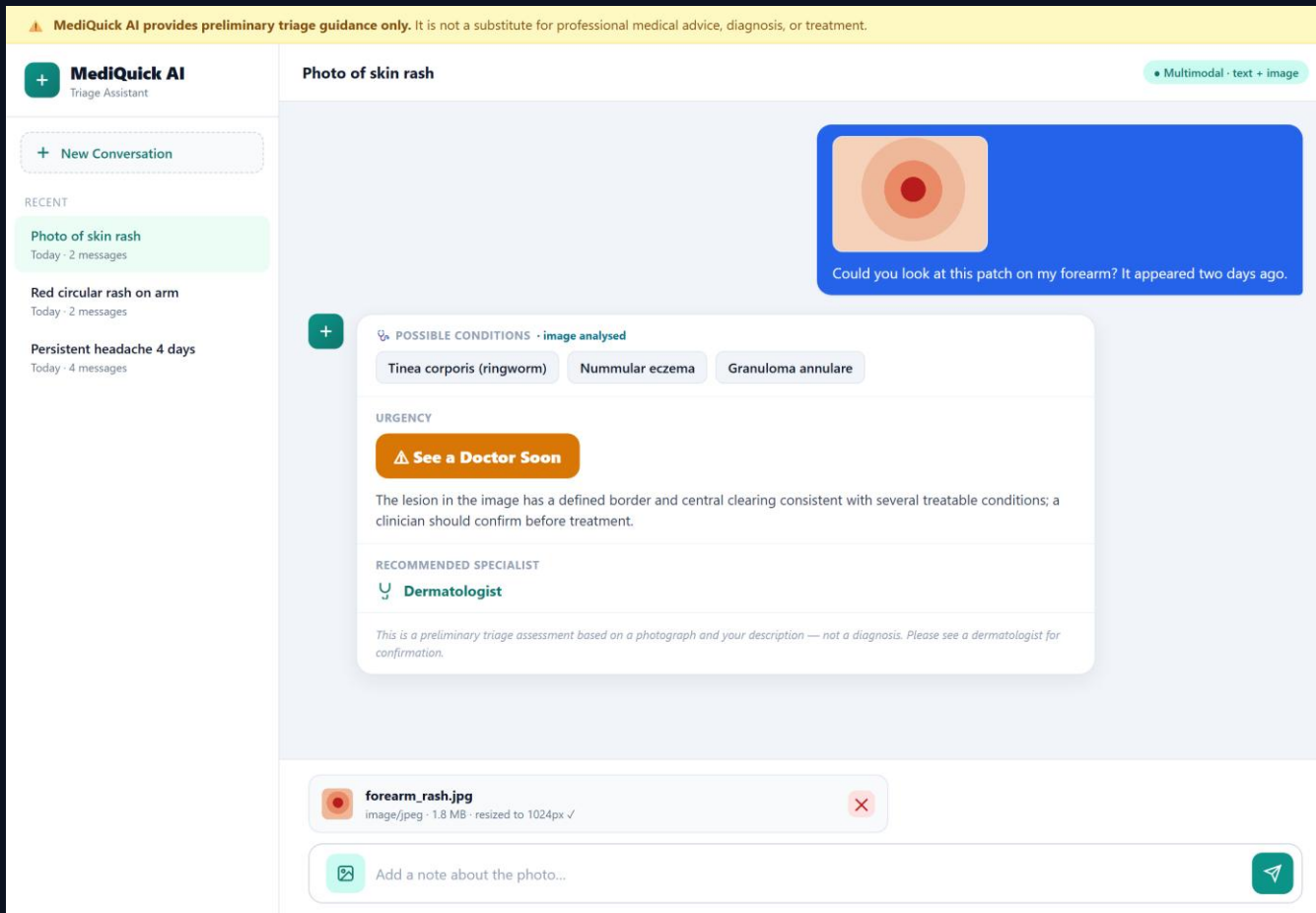
- Have you recently been in contact with any new substances or products?
- Have you traveled to any areas with high grass or vegetation?
- Have you experienced any fever or swelling around the rash?

This is a preliminary triage assessment and not a definitive diagnosis. Please consult a doctor for a thorough evaluation and proper diagnosis.

Describe your symptoms... (Shift+Enter for a new line)

Triage card: conditions, urgency, specialist, and follow-up questions (prototype UI).

+ Demo • Photo Analysis



A photo is validated, resized, and analysed alongside the text — in memory only.

In-memory

Photos are never written to disk

≤1024 px

Resized & JPEG-encoded for the model

text + image

Analysed together, one assessment



Validated Live · All Three Urgency Tiers

MediQuick AI — Structured Triage Across All Three Urgency Tiers

Live, unedited output from the backend calling NVIDIA NIM (Llama 3.2 90B Vision), 19 June 2026.

"I have a tiny shallow paper cut on my finger, no bleeding now, just a little stinging."

POSSIBLE CONDITIONS

Minor cut / laceration Paper cut Skin irritation

URGENCY

✓ Manageable at Home

The injury is minor and not bleeding, suggesting no immediate medical attention is required.

SPECIALIST

General Physician

"A red circular rash on my arm for 3 days, slightly itchy with a clearer center."

POSSIBLE CONDITIONS

Ringworm Contact dermatitis Eczema

URGENCY

⚠ See a Doctor Soon

The rash has been present for 3 days and is itchy, which may require further evaluation and guidance from a doctor.

SPECIALIST

Dermatologist

"Sudden crushing chest pain spreading to my left arm and jaw, sweating and short of breath."

POSSIBLE CONDITIONS

Acute Coronary Syndrome Myocardial Infarction

Angina

URGENCY

🚑 Seek Emergency Care

Sudden crushing chest pain, shortness of breath, and sweating are symptoms that require immediate medical attention.

SPECIALIST

Emergency Medicine / Cardiologist

self_care

consult_doctor

seek_emergency

The emergency override correctly escalated the chest-pain case regardless of other factors.

Real, unedited output from Llama 3.2 90B Vision — the emergency override correctly escalated the chest-pain case.



Progress at the Milestone

46%

of the project complete

6 of 13 modules — the entire backend

Completed

- ✓ FastAPI backend + SQLite
- ✓ NVIDIA NIM client (text + vision)
- ✓ Prompt engine + JSON parser
- ✓ Multi-turn context manager
- ✓ Image pipeline + /upload
- ✓ /chat & /sessions endpoints
- ✓ All 3 tiers validated live

Remaining

- React chat interface
- Triage card + urgency badges
- Image-upload component
- Session sidebar & history
- Disclaimer banner (UI)
- 30-scenario evaluation CSV
- Polish, responsive, README

What's Next

1

Build the React interface Chat window, colour-coded triage card, image upload, session sidebar.

2

Complete the safety layer Persistent on-screen medical disclaimer banner.

3

Run the evaluation 30 scenarios across three tiers → accuracy CSV (target $\geq 70\%$).

4

Polish & hand over Error toasts, responsive layout, and a developer README.



Thank You

MediQuick AI provides preliminary triage guidance only — not a medical diagnosis.

